

PODCAST

Hitting the Buffers: The 1873 railway bust that broke one of America's greatest financiers

The Story of Money

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HIGH RELEVANCE

The episode is an extended historical analogy for the current AI infrastructure capex cycle, directly relevant to JM's holdings in NVDA and MSFT and to second-order effects on energy, infrastructure and cyclical names across the portfolio.

TL;DR

The 1873 Panic was the collapse of a debt-fuelled infrastructure mania that was, in relative terms, roughly ten times larger than today's AI capex splurge as a share of GDP. The railways themselves survived and eventually made America rich; the bond investors, the lead underwriter and six years of the broader economy did not. The episode's core argument is that infrastructure bubbles often leave behind genuinely useful assets, but that the financiers and early-cycle capex suppliers absorb the losses while late-cycle users and survivors capture the value.

What it means for my portfolio

NVDA NVIDIA Corporation · Held · Direct

CHALLENGES

NVDA is the closest present-day analogue to Jay Cooke: the indispensable enabler of the mania, whose fortunes are tied to continued capex issuance by others. The episode's central lesson is that being the dominant financier or supplier of a genuine technology revolution does not protect you from the bust — Cooke was right about railways and still went bankrupt. The stranded-asset risk (LLM architecture displaced by something more efficient) and the contagion-trigger risk (foreign liquidation of US tech) are both live. JM has no formal thesis document for NVDA in the provided context, so there are no named pillars to assess, but the holding warrants explicit scenario-testing against a 30-50% revenue correction.

ACTION · REVIEW THESIS

MSFT Microsoft Corporation · Held · Indirect

SUPPORTS

Microsoft is better positioned than NVDA in the railway analogy: it is more like one of the larger, better-capitalised railway companies that survived bankruptcy and emerged with cleaned-up balance sheets, rather than the bond underwriter who went down with the issuance. Its diversified revenue base (Office, Azure non-AI, gaming, LinkedIn) provides the equivalent of operating cash flow that the pure-play railway bond houses lacked. The risk is that Azure AI capex commitments create a balance-sheet drag if utilisation disappoints, but this is manageable at Microsoft's scale. No formal thesis pillars provided in context.

ACTION · WATCH

WDS Woodside Energy Group · Held · Thematic

SUPPORTS

PILLAR P1

The episode explicitly flags energy infrastructure as the most likely durable residual of the AI capex boom — power generation capacity being built to feed data centres will remain useful regardless of which AI architecture wins. This supports the broader thesis that energy infrastructure is undervalued relative to the digital layer it enables. For Woodside specifically, LNG demand as a power-generation fuel in data-centre-heavy geographies (US, Southeast Asia) is a secondary tailwind, supporting P1's assertion that the infrastructure value is what matters.

ACTION · NO ACTION

NHC New Hope Corporation · Held · Thematic

SUPPORTS

PILLAR P1

AI data-centre power demand is a genuine incremental source of thermal coal and gas demand in markets that cannot quickly build enough renewables — this is a second-order support for NHC's P1 thesis that coal demand outlives the political consensus. The analogy is indirect but not forced: iron foundries boomed during the railway construction phase; thermal coal is a foundry-equivalent input in the AI power-demand phase.

ACTION · NO ACTION

BXB Brambles · Held · Thematic

NEUTRAL

No material link. Brambles is a keepsake holding with no exposure to the AI capex cycle or the historical dynamics discussed.

ACTION · NO ACTION

SO WHAT

The 1873 episode is a genuine and useful historical lens for JM's two largest thematic exposures — NVDA and the AI capex cycle more broadly. The core lesson is not that the technology is wrong but that the dominant enabler of a mania (Cooke, and by analogy NVDA) is the most exposed name when sentiment turns, even when the underlying technology proves transformational. JM should ask whether NVDA's current price embeds a scenario where AI capex growth merely slows to normal rates rather than contracts, and stress-test the holding against a 40% revenue correction. The durable residuals of the AI boom — energy infrastructure, grid capacity, and eventually the hyperscalers who survive with cleaned-up balance sheets — are where patient capital should be positioned for the decade after the bust, not the decade before it.

3 TO 5 YEARS

The medium-term lens

Themes

- AI infrastructure as a debt-fuelled capex mania with a historical precedent an order of magnitude larger, suggesting the bubble need not be near its end but that the eventual bust will be felt unevenly across the supply chain
- Stranded-asset risk in data-centre and semiconductor capex if LLM architecture is supplanted, analogous to iron foundries that closed when railway bond demand evaporated
- Energy infrastructure as the potentially durable residual of the AI boom — the speakers explicitly flag power generation capacity as a lasting beneficiary regardless of which AI model wins
- Contagion transmission: a financial shock in a seemingly unrelated geography (Austria 1873, today potentially China property or European bank stress) can be the pin that pops a domestic bubble already stretched by illiquidity

Historical parallels

THE PARALLEL

1873 railway bond mania vs 2024-25 AI capex cycle: both involve a transformational technology, mass retail and institutional participation in the funding instrument (bonds then, equities and private credit now), a dominant underwriter/enabler (Jay Cooke then, NVDA now) whose fortunes are tied to continued issuance, and government adjacency that creates moral-hazard assumptions about backstops. The analogy holds on the structure of the mania, the role of leverage, and the 'carnival barker' dynamic in financial marketing.

WHERE IT BREAKS

The 1873 mania was roughly 10x larger relative to GDP, was financed almost entirely by bonds rather than equity, had no central bank to provide a liquidity floor, and operated under the gold standard so deflation was the automatic adjustment mechanism. Today the Fed can and does intervene; AI capex is largely equity- and cash-funded by companies with strong balance sheets (Microsoft, Google, Amazon); and the sovereign backstop is implicit. The bust, if it comes, is likely to be slower and more contained, though not absent.

THE PARALLEL

1990s fibre-optic/telecom bubble vs AI data-centre build-out: both involve massive physical infrastructure investment predicated on demand forecasts that may prove optimistic, both leave behind assets that eventually get used (fibre became the internet backbone; data centres may become general compute infrastructure), and both see the primary capex suppliers (Cisco then, NVDA now) trade at multiples that embed years of uninterrupted demand growth.

WHERE IT BREAKS

Fibre was a commodity once laid; NVDA's chips depreciate, require power, and may become obsolete if inference efficiency improvements reduce the number of chips needed per unit of output. The asset durability argument is weaker for semiconductors than for glass fibre in the ground.

Second-order effects

- If the AI capex cycle turns, the first casualties are the capex suppliers (semiconductors, networking, server OEMs), but the second wave hits the energy infrastructure builders who are now committing to long-dated power contracts and generation assets on the assumption that data-centre load is permanent — a risk for any energy utility or developer currently repricing on AI demand.
- A contagion event from outside the US (European banking stress, China property second leg, or an EM currency crisis) could be the Austrian butterfly that forces liquidation of US tech equities by foreign holders, since non-US investors are now

heavily exposed to US large-cap tech through index funds — exactly the channel that broke Jay Cooke.

- The 'good bubble' residual — durable energy infrastructure, fibre, and general-purpose compute — benefits patient capital that buys the survivors after the bust at distressed prices, not the capital that funded the boom. JM's 'slow to buy, even slower to sell' philosophy is structurally well-suited to that second phase, not the first.
- Iron foundries in 1874 are the analogue for today's commodity suppliers whose order books are swelled by AI-linked demand: copper, power transformers, cooling systems. If data-centre build slows sharply, these suppliers face the same demand cliff the foundries did.

Winners and losers

In the railway analogy, the infrastructure itself (track, bridges, terminals) survived and eventually created enormous value for users — farmers, manufacturers, consumers — and for the railways that emerged from bankruptcy with cleaned-up balance sheets and no legacy bond coupon. The losers were the bond investors, the lead underwriter (Cooke), and the broader economy for six years. Translating this forward: if the AI capex cycle corrects, the likely survivors and ultimate value capturers are the hyperscalers with the strongest balance sheets (Microsoft, Amazon, Google) who can absorb write-downs and continue operating, and the energy infrastructure that gets built regardless of which AI model wins. The likely losers are leveraged data-centre developers, the semiconductor supply chain at peak-cycle multiples, and any company whose revenue model depends on continued AI capex growth rather than AI adoption by end-users. NVDA, as the Jay Cooke of this cycle — the indispensable financier of the mania — is the name most exposed to a sentiment reversal even if its technology remains genuinely superior.

THE EPISODE

What was said

- By 1874, US railway bonds outstanding reached US\$2.2 billion; by 1890, US\$5 billion, equivalent to roughly a third of US GDP that year — the speakers calculate this implies a US\$10 trillion capex boom in today's terms, versus AI data-centre capex currently running at roughly 1.2% of GDP.
- Jay Cooke agreed to underwrite US\$100 million of Northern Pacific bonds at 7.3% interest, buying them from the company at a 12% discount, while advancing cash that swelled from a promised US\$500,000 to US\$7 million as bonds went unsold — a classic underwriter's liquidity trap.

- The trigger for the 1873 crash was the May 1873 Austrian Gründerkrach (entrepreneurial crash), which caused European investors to liquidate American bond holdings — demonstrating that a geographically remote financial shock can detonate a domestic bubble already under strain.
- The New York Stock Exchange closed for ten days and lost a quarter of its value; by end-1875, over 120 railways with US\$550 million in bonds had defaulted; half of all US iron foundries closed by end-1874; corporate bankruptcies peaked at 10,000 in 1878.
- The episode was so severe it was called 'the Great Depression' until the 1930s episode displaced it; the recovery took six years, and the period is now known as 'the Long Depression'.
- Jay Cooke held up to 60% of Northern Pacific's equity but did not control the board — he had the downside of a principal and the information of a passive minority, a structural error the speakers identify as a key reason he could not correct course.
- The speakers note explicitly that not all capex booms are equal: railway track still sits in the ground 150 years later, whereas data centres built for large-language-model inference may be stranded if a different AI architecture wins — the VHS/Betamax risk.
- Robin Wigglesworth (author, FT) is writing a book on bond market history; Gillian Tett is FT editor-at-large — neither is a fund manager talking a book, though both have reputational incentives to make history sound relevant to today.

TEST THE STORY

Sceptic's corner

Wigglesworth is writing a book on bond market history, so framing the 1873 railway bust as the most spectacular debt-fuelled capex splurge in history serves his narrative. The comparison to AI capex is made without fully acknowledging that today's primary funding instrument is equity and corporate cash, not debt — the credit-crisis transmission mechanism that destroyed Jay Cooke (illiquid bond inventory, no lender of last resort) is materially different from a scenario where Microsoft or Google simply write down overbuilt data centres on a strong balance sheet. The speakers also gloss over the fact that the railways required physical land, political violence, and lethal construction conditions that have no analogue in data-centre construction — the frictions that made the 1873 bust so severe were partly physical and jurisdictional, not purely financial. Finally, the 'good bubble' argument (bubbles leave useful infrastructure behind) is presented optimistically but without rigour: the six-year depression that followed 1873 killed real people and real businesses, and the infrastructure benefit accrued mostly to later generations. Investors in the mania did not get to enjoy it.

New ideas

- Power generation and grid infrastructure companies (not currently held or watched): the episode and the speakers both argue that energy capacity built for AI data centres is the most durable residual of the current mania — worth researching ASX-listed or globally listed utilities and grid operators with contracted data-centre load, as these are the equivalent of the towns that grew up around railway depots
- Post-bust semiconductor or data-centre real-estate plays: if the AI capex cycle corrects, the companies that buy distressed assets at cleaned-up prices (as railways re-emerged from bankruptcy) may be the best entry point — worth building a watchlist of data-centre REITs and fabless semiconductor designers trading well below replacement cost, for purchase after a correction rather than before

REFERENCE

Companies mentioned

COMPANY	TICKER	CONTEXT
Jay Cooke & Co		The failed 19th-century bank at the centre of the 1873 Panic; the episode's primary subject
Northern Pacific Railway		The railway Jay Cooke underwrote and was financially destroyed by
Union Pacific	UNP	One of the two companies building the first transcontinental railway, financed by government bond guarantees and land grants
Central Pacific		The other company building the first transcontinental railway from the west coast
Credit Mobilier of America		Sham construction company used to siphon federal subsidies from Union Pacific; one of the biggest scandals in US political history

FOLLOW UP

What I'm watching next

- ☐ NVDA quarterly revenue growth rate and data-centre segment as a share of total — any deceleration below 30% year-on-year for two consecutive quarters would warrant urgent thesis review
- ☐ Hyperscaler capex guidance (Microsoft, Google, Amazon, Meta) at their next quarterly results — if any one of them signals a pause or reduction in data-centre build, the contagion to NVDA and the broader capex chain will be fast
- ☐ Foreign holdings of US tech equities via TIC data or ETF flow data — European or Asian liquidation of US large-cap tech is the Austrian-butterfly channel identified in the episode
- ☐ Power generation and grid infrastructure order books at companies like Schneider Electric, Eaton, or ABB as a leading indicator of whether the energy-infrastructure residual thesis is building momentum
- ☐ Any sign of an alternative AI architecture gaining traction at scale (world models, symbolic AI, efficient inference) — this is the VHS/Betamax risk the speakers name explicitly, and it is the kill condition for data-centre capex durability

Full transcript

0:00 Jay Cook & Co had three main branches. In New York, he had his own partners that were very able, but they literally just ran out of money. They didn't have enough money to meet their own obligations. So the New York partners summon all the big counterparties among the big brokerage houses and investment banks on Wall Street and say, look, if we don't have a million dollars by 10 a.m., we're dead. Wow. Yeah.

That's where you get the spooky music and dramatic music into the movie. It is the moment, really. It electrified everybody. This was not a small bank. This was like a J.P. Morgan or a Goldman Sachs out of the blue, suddenly saying, we're dead.

Today on the story of money, what happens when an exciting new technology fuels an investment bubble, attracting a huge amount of money that eventually stages a spectacular burst. Now, we could be talking about the present day AI boom, but in fact, we're going to bring you a story that took place 150 years ago. It's about the technology that first knitted the US economy together, financed by an unprecedented accumulation of debt. Vast fortunes were made as the market built up a head of steam before hitting the buffers in one of the biggest financial panics in Wall Street history. One that derailed the US economy for years afterwards. So if you still haven't figured out which technology we're talking about,

well, Robin, do you want to do the honours? It's the railways, of course. Of course it is. And you care a lot about railways, don't you? Well, the huge railway bond boom of the late 19th century is just a fascinating episode. It's perhaps the most spectacular debt-fueled capital expenditure splurge in history.

Tens of thousands of miles of track was laid across the US from east to west, north to south, and really kind of set the path to the US becoming the world's mightiest economy. And in many ways, the railway boom was also proof that you can have an amazing revolution technology that seems good, but it doesn't mean that it's going to be a straight path forward for finance. On the contrary, you can have some very damaging booms and busts, and that's got a lot of lessons for today's AI mania, which we'll be looking at in today's episode. But Robin, you're the one who's going to be talking about this most of all, because you've written a book on it, haven't you? Well, it's a chapter in a book. I'm writing a book on the history of the bond market,

and there's a chapter on the railway bond boom of the 19th century because it's spectacular. It really is just a glorious, demented period of finance and economic history, really. It was when railways really kind of knitted physically the United States together. It's a bit like the internet today in cyberspace, but railways did it in physical ways. So Alexander Hamilton first issued treasury bonds to kind of knit those first colonies together into one federal state. Railways kind of physically brought the United States.

3:03 It made the United States in, I'd argue, a very real sense. It also helped spark the first great, big corporate bond market. There was a massive moment in the history of finance, and that's what we're going to be diving into today. So for all those reasons, you need to listen on. This is me, Gillian Tedd. And me, Robin Wigglesworth, on The Story of Money.

So Robin, we know how this story is going to end. It's a crazy boom that turns into a bust. It involves the railways. It involves manias. It's got parallels with Lehman Brothers. It's really dramatic.

But before we start that, and before we start talking about the parallels with, say, the dot-com bubble or 2007, or even the AI world today, explain to us how and why this crazy bubble formed, and give us a sense of what happened on the moment it actually went bust. So it was actually called the Panic of 1873 for a long time. It happened on 18th of September, and it was the culmination of a just, frankly, epic eight-year-long construction boom in railways that came after the U.S. Civil War. And it was all financed by bonds, really. And it was an epic crash on the stock market, actually.

The New York Stock Exchange had to close for 10 days. It was the longest closure ever in the New York Stock Exchange up until the outbreak of World War I. The exchange lost a quarter of its value, actually. It was, you know, if not quite Black Monday, it was pretty close to it. It was a big one. So I actually have a report here from the New York Times from that day that actually really captures it, and it's vivid, so I'm gonna read it out to you.

The brokers stood perfectly thunderstruck for a moment. Prices were declining frightfully. Some of the men who were ruined swore. Some of them wept. Some went out to the street without saying a word. Others talked of the trouble in a jovial way and went about trying to borrow money from friends.

That's from the New York Times the day after the panic of 1873. I guess it's fascinating because you talk about the bond markets. It wasn't just the stock markets. Yeah, so the bond markets is kind of really the epicenter of this crisis. But in the crash itself, the real focus were the banks themselves. In the 19th century, we often call financial crises panics because there were literally panics.

People were worried about the safety of banks. There was no Federal Reserve in the United States. There was no deposit insurance. If the banks were unsafe, well, you went there and yanked your money out in a panic. And that, of course, would cause the bank to collapse. And it would typically cause a domino of banks to collapse.

So quite literally, people were running the bank or queuing up. Exactly. And the bank at the center in 1873 was something called J. Cook & Company. That was the bank that collapsed that day and caused that frightful reaction that the New York Times detailed. So a bit like the Lehman Brothers of the 19th century.

6:11 And at the heart of this whole dramatic story, there was J. Cook & Company, which isn't that well-known today. Yeah, so J. Cook was the man. I mean, even calling him the John Pierpont Morgan of his day actually underplays how titanic he was. He was the world's most powerful financier.

He was even more powerful than the Rothschilds, arguably. His bank was synonymous with American finance, almost synonymous with American economic might at the time. So tell me, I mean, I've never heard of J. Cook. I clearly should have done. Where did he come from and how did he get so rich?

Well, so he was born in rural Ohio when Ohio really was the frontier state. He came from a relatively affluent but unremarkable family. His father had a really weird name, Eleftherios or something. It was named after a really obscure Greek saint and he hated his name so much he gave all of his sons very simple names like J, John. So J. Cook became kind of a financier almost by accident.

He ended up being apprentice to a local bank and he did really well. He made partner when he was still in his twenties and took over the bank in his thirties. Bonds were his metier. He's a brilliant bond salesman, bronze structurer. And actually he was so good that he retired in his mid thirties because he'd basically achieved everything he wanted to achieve, he was wealthy. So was he a bit of a slimy Weasley kind of entrepreneur or was he more like a classic 19th century big sort of robber baron figure?

I mean, what kind of person was he? Neither really, he's a man of immense contradictions. He looked a bit like Santa Claus. If Santa Claus looked really mean, he had big glowering eyes, a big white bushy beard. He was incredibly intense. He was not like a chit-chatty guy.

He's not a slick salesman but he was clearly incredibly brilliant. He really admired Native Americans. He named his estate later on after a chief from the Weandot tribe that he really had met several times when he was a child, used to run on his shoulders and just admired hugely. But the railways he financed and ran later on, frankly, just ripped apart huge parts of Native American lands. He was an ardent abolitionist but he wouldn't let people of color ride on his tram lines. He hated graft and bribery.

He said it at least, but he was pretty good at it. He didn't like corruption but he was very good at cultivating people in power. And he was also fabulously wealthy but he lived a pretty simple Puritan life apart from this massive estate that he built called Ogontz after that chief I mentioned. So he came from Ohio. Did he carry on

living in Ohio? No, by then he'd moved to Philadelphia which was arguably maybe the financial center rival to New York at the time.

9:04 So he does sound like a really fascinating character. And as you say, somebody who actually epitomized a lot of the contradictions, not just in finance but actually in the American country and nation as a whole back then when you had both this very high-minded sense of biblical pioneer purpose coupled with this wild entrepreneurialism and get-rich-quick mentality in many ways. No, I think so. He's kind of in many ways a quintessential American financier. That sadly we have forgotten a bit today but he was just a fascinating character. I'm not saying he was an amazing human being because he had amazing science but he also had darker science clearly.

And I think that makes him even more compelling at least to me. Right, it's pretty that Tocqueville didn't meet him to write about him. Yes, exactly. It's a century too late, I think. So we have this man who's a complete contradiction who's a rags to riches story and then bringing me back to rags again but also in many ways epitomizes the contradictions in America. But I'm curious, he starts off on rural Ohio.

How on earth did he get involved in the railway boom and high finance? Well, he moved to Philadelphia in 1839. Philadelphia was at the time a major financial center as well. His first big moment was arguably selling bonds on behalf of the US government to finance its war against Mexico in the 1840s. It always helps to get close to the American government. It does, doesn't it?

I'd say this reflected more his skill as a bond salesman. He had sold lots of bonds on behalf of railways already, canals, factories, businesses in the US. Bonds were a big thing already in the United States. Was he just a kind of a natural salesman or did he actually come up with some clever innovations as well? The main thing we saw that was in the Civil War when really he rose from being sort of probably one of the preeminent bankers in America to being a titan. He came kind of out of retirement to help the North raise money to finance the war against the South.

Because initially the war was not going well for the North. It was, especially with Philadelphia, was struggling to sell bonds to finance its contribution in the fight. And he basically stepped in and just did an incredible job essentially. So it sounds like, you know, as ever wars don't only spark a lot of innovation, they also enable some people to make fortunes. And he was clearly one of those. So why was he so good at selling bonds?

Well, the main innovation again, that we really saw in the Civil War was mass selling. You know, bonds were typically sold to banks, to big insurance companies at the time, very wealthy individuals, maybe a few sort of relatively well-to-do shopkeepers and people like that. But he realized that the North had to mobilize entire citizenship. This was kind of what Clausewitz would call total war beyond the financial sphere. So he sold bonds in extremely low denominations, so those \$50, but you could also slice it up into different ways so that anybody could and should join in to fight. Even if they couldn't send, you know, their sons and daughters and so on to fight, they could help the war effort by buying the North's bonds.

12:17 He marketed, he sold, he handed out leaflets. He did these little explanations and Q&As. They put in all the newspapers in the world. It was incredibly aggressive. It's kind of the first big bond push that we actually saw the U.S. replicate almost verbatim in World War I with the Liberty Loan Program.

And that's how he managed to do it. And it was a big deal. Even Ulysses Grant said that, you know, once they finally had North at one, that, you know, send somebody to cook and say, tell him that it is to his labors more than those of any other man that the people of this country owe the continued life of the nation. So Ulysses Grant, of course, was a general of the Union Army. And to get that kind of praise is very striking. How did Jay Cooke react to that?

Well, it probably won't shock you that he was not a fantastically humble man. He did not downplay this. I've got another great quote from him. In his own words, like Moses and Washington and Lincoln and Grant, I have been, I firmly believe, God's chosen instrument, especially in the financial work of saving the Union. Well, the

hard fact is that many financial geniuses and big titans of Wall Street and the City of London also have a bit of a Messiah God-like complex as well. Touche.

Exactly. So we have this extraordinary figure who has done amazingly well, not just originally in the bank, but then got on to supposedly single-handedly save the Civil War for the North by selling bonds in this mass market way. Why on earth doesn't he just, at that point, go off and spend his time fishing and having a nice, quiet life? Why does he get involved in railways? I guess a bit like it's like Al Pacino in the third Godfather film, where he says, like, I tried to leave, but they pulled me back in. But really, I think it's his ambitions that pulled him back in.

And this is 1865, isn't it? Yeah, exactly. When the Civil War is now over. Yeah. And basically they're about to start the task of trying to rebuild the country. Yeah.

So there'd been lots of talk of transcontinental railways before the Civil War, but this became kind of the, for political and cultural reasons, the great unifying project of America after the Civil War, because there were lots of unemployed soldiers, the country's economy was a shambles, and they thought that transcontinental railways could be this, you know, in very real practical way, a valuable thing, but also symbolically powerful in tying the country together. The pre-war network of railways was around 25,000 miles of track. That was basically all in the states east of the Mississippi. That was a lot of railways. I mean, there was as much as we had in Europe, but it wasn't nearly what it was about to become.

15:00 The government decided it was gonna finance with bond guarantees and land grants the construction of the first transcontinental railway by two companies, the Central Pacific and Union Pacific. And they were building from each coast and they were gonna meet together somewhere in the middle. Wow. Well, this sounds like an extraordinarily ambitious vision, not just technologically, but also in terms of nation building and, of course, financially as well. I mean, who were the people who were actually building this? Was this a period of Irish immigrants or were there others involved as well?

Yes, so there were lots of unemployed soldiers, but a lot of Irish laborers and Chinese laborers. Chinese as well? Gosh. Yes, a lot of Chinese laborers. They were on the first base of booms there. Someone should tell Donald Trump that Chinese helped to build America.

No, exactly. They helped literally build the physical skeleton of the American economy. They were huge contributors. And this is not glamorous or easy work. This was incredibly hard in flat stretches of land close to civilization, but in the middle of nowhere, in mountains and gorges and prairies and deserts, it's lethal work. I mean, thousands of people died building these railways.

And so just like in the way the internet completely transformed our imagination of how we all communicate with each other and interact with each other and made businesses possible where there were no businesses before, I'd imagine the railway must have done the same thing as well in the middle of America in terms of towns or industrial complexes and things. Well, in Europe, railways typically connected towns together, towns and villages and cities in different countries. In the US, railways literally created towns and cities. There were towns and cities that built up around where they built the railways. They were kind of even more sort of risky venture capital type investments, a bit like today with AI,

than they were in Europe. These were hugely expensive, incredibly dangerous and highly financially risky projects. And one of the things, of course, that the internet did was to really crush the cost of communication. Did the same thing happen with putting the railways in in terms of the physical movement of people across the country or goods? Yeah, before there was almost no way or no easy way of going from east coast to west coast. So this was a country that ran, you know, north to south, south to north on the east coast.

The trains just revolutionized that, completely changed the axis of the entire country and then massively, radically lowered the cost of transportation. So you could, in theory, after the transcontinental railways were built, you could eat Maine lobster in Los Angeles and you could have Californian oranges in New York

because it was so much easy and cheap and quicker to transport these goods. So in many ways, it was a kind of win-win-win in the sense you were essentially stitching the country together to create a sense of nationhood and presumably driving lots of growth as well. What's not to like? Yeah, the railways turned the US into a proper industrial country, a country that had arguably been agrarian

18:05 or agricultural beforehand, became an industrial powerhouse thanks to railways. And of course, made some people very rich too. Yes, I think, you know, what is sometimes forgotten about the railway boom is that, you know, the financial tracks it ran on was not the stock market, even though that gets a lot of attention, it's actually bonds. The amount of bonds issued by towns and railway companies to build these things was just unimaginable. So I've got some numbers here that really sort of hammers home the scale. By 1867, the US railways had issued \$416 million worth of bonds, which was quite a bit for the era, but not massive.

But by 1874, it had reached \$2.2 billion, and that was a massive amount at the time. And by 1890, that was \$5 billion worth of bonds. And what is that in today's money? Well, so if you just adjust for inflation, it's around \$180 billion, which is a lot, but again, doesn't sound that much. But you have to realise that the US economy was so much smaller back then. So GDP in the US in 1890, let's take that year, was around \$15 billion.

So \$5 billion worth of bonds equates to basically a third of GDP. So if we translate to today, we're talking a \$10 trillion capex boom in one industry. So you're saying in relative terms, the railway boom involved more finance than even the AI boom today? Yeah, it is. The AI data centre splurge, the building of it, is now, it seems, by most data points we've looked at, the biggest since the railways. But it's around 1.2% of GDP.

The railway boom, we're talking some 10, 15% of GDP. So AI maniacs eat your heart out. Yeah. And how does this compare to, say, government spending? Because normally we think the governments do most spending, don't we? Well, this was a prior.

private market as well. That's what's so interesting. Yes, bonds primarily had been a government financing tool. The US had borrowed a lot to finance the Civil War. Obviously, that's fairly expensive, but even that was dwarfed. So, it was around \$2 billion worth that the US government borrowed, the northern government borrowed to pay for its efforts to keep the country together. So, the railway boom was more than twice that.

It's just head spinning. I mean, and it raises the question though, if you're going to issue these bonds and need all this finance, who the heck is going to finance it? Where did all the money come for this massive issuance? Who were the buyers? So, it's important to remember that this came after the Civil War. And actually, those techniques that Jay Cook had pioneered during the Civil War to finance that \$2 billion bond splurge was replicated by him, but also everybody else subsequently. This was basically an entire

population that had been primed to buy bonds. They'd been educated about bonds for years and years and bought lots of bonds, and frankly, probably made a decent amount of money by lending money to the US government. And that obviously whets the appetite. If you made a bit of money, you want to make more money. And then afterwards, you have these incredibly compelling visions of this hot new technology. So, it's actually pretty easy to sell to people.

21:21 There's a great quote from the historian Richard White, who's written one of the best books on this episode, which shows how the progressive railway boom grew out of the Civil War bond boom. Here it is. Investors proceeded from government bonds to government-secured railway bonds, to convertible bonds, to mortgage bonds vouched for by the same people who sold the government bonds, to a whole array of financial instruments, and from there, potentially into the drink. There were first mortgage bonds and second mortgage bonds. There were mortgage bonds on the trunk line and mortgages on the branches. There were land-grant bonds and income bonds.

There were bonds on anything and everything that investors might accept as collateral. The bankers, as well as the railroads they represented, trumpeted the security of these investments. They were, in effect, so many carnival barkers. Wow. Yeah. And for a long time, it works.

It always does. Everyone makes money. It seems to work. Exactly. So, you've got this wild mania around every type of bond that moves. And, in fact, you've got these wonderful phrases like carnival barkers, which I love. Was Jay Cook a carnival

barker? Where did he fit into all this? No, I mean, like I said, he was quite an austere guy. He was very clear about differentiating speculation from investment. And he had invested and helped back railway bonds before the Civil War and a few afterwards, but solid, serious ones back between towns and had a clear economic rationale. He did not initially believe in the mania for transcontinental railways because after one was built, people wanted to build more and more and more. This was the hot new

thing. And he actually, to his credit, did stay out of it for a little bit. Right. Well, good for him. How close was he to the government? Because the government must be involved somewhere in the background in all this, was it?

Yes. Cook was reasonably close to the government. He always stressed that he didn't like corruption and graft and nepotism, but his brother, especially, was extremely influential in the Republican Party. And Jay Cook definitely benefited from that. To his credit, for example, when he sold bonds on behalf of the US government during the Civil War, he charged extremely little. His commission was very small. And when there was an outcry against that, he cut it further.

So he did not sort of become a wealthy man for stiffing the US government. But he definitely benefited from ties both directly with him and his brother to the Republicans. And then later on, Ulysses Grant became President Grant as well. And they were certainly very close. Well, I guess if that had been 2025, 26, there would have been photographs of him at the inauguration in the White House for the new president, wouldn't there? Oh, definitely. Well, President Grant actually came to a house party, several house party

24:19 at Jay Cook's Organs Mansion house, including his inauguration. So just to underscore the importance of Cook, Grant came to his housewarming party. I don't know if Cook came to Grant's housewarming party at the White House. Ha, well, that's kind of funny, because it then shows that all the handwriting about the links between the AI and big tech sector in the Washington Day is part of a long established historical tradition. But I'm curious, because, you know, Ulysses Grant, insofar as I remember my history books, wasn't quite the same as Washington and Eisenhower, was he? I mean, he's often seen as being one of the more corrupt presidents that America's had.

To be fair to Grant, it's unclear whether he was personally corrupt. But despite being a great wartime general, he seemed to have been an incredibly, incredibly bad judge of character. There's literally a Wikipedia page just with all the scandals in his administration, which I don't think is a good thing. And of all the scandals that developed at the Grant presidency, the biggest by far was a credit mobilier scandal. So that involved the Union Pacific Railway executives creating this entirely sham construction company called Credit Mobilier of America, and they used it to siphon off federal subsidies and profits. And this scheme predated Grant's presidency, but it involved a lot of Republicans close to him,

including his vice president, Skyler Colfax. So it still is today probably one of the biggest scandals in American history. And that's a lot of competition. Yes, very true. So you have the ingredients for this really heady cocktail of a president who is a war hero, but presiding over a corrupt administration. And you have this extraordinary character, Jay Cooke, who says he's incredibly biblical and upright, but is also very ambitious and busy creating all kinds of new financial marketing schemes around bonds and want to build railways. So let's stop there for a commercial

break. And when we return, we'll talk about how this story starts to go horribly wrong around the North Pacific Railway. Okay, we're back in 1869. And it's a pretty momentous year, because the Civil War is over. The new railway boom is underway. It's bigger than anything America's seen before. And

27:02 essentially, they're using this astonishing race to build the railways to knit the country together, create jobs, create an economic boom, and of course, a financial boom too. But as so often in history, what starts off as a good idea is starting to go bad, because there's money being funneled into not just good schemes, but bad ones too. So tell us how Jay Cooke enters the story at this point, because he was also something of a hero at this point. He had done a remarkable job in financing the Civil War and having a stellar career. And yet, something starts to go wrong. Yeah, he was fabulously wealthy, enormously successful. And to his credit, initially,

he stayed out of the whole kind of railway mania, bond mania that started after the Civil War until 1869. And that's when he makes a horrible, but no good, terrible mistake. But he could have just gone fishing. I mean, he was already super wealthy. He had nothing to prove. Yeah.

Why would he get suckered in? Ironically, he loved fishing. That's exactly what he should have done. His only two hobbies were fishing and discussing theology with seminarians and priests. Well, memo to all you financially successful financiers, know when to stop. Get out.

Exactly. You can't go wrong with just going fishing, basically. So what actually happened? You have to remember, this is a guy with a titanic ego at this stage. And the big economic story at the time are the railways. And that is his metier. I mean, he made his name first by

selling railway bonds, and he could see everybody else get rich around him. Maybe not Jay Cooke rich, but they're getting pretty rich. And he thinks, well, I know this better than everybody else. So there was a plan for a second transcontinental railway on the Northern Plains from Montana, Seattle, that had failed to raise financing. It was super risky. They'd already been the first one, and this was obviously then going to be even riskier.

And Cooke had twice before said no to getting involved. So like, this is not for him. This is speculation. I don't do speculation, essentially. But he gets sucked in. He, for some reason, he changes his mind. And I think it's just this sense of wanting to be

the big man. But couldn't he gone off and done politics instead? Because that's what a lot of big Wall Street figures in the last few decades have done. When they've achieved everything in finance, they go off and try and become treasury secretary or central bank governor or something like that. Yes. I think one of the big sliding door moments of this story is that he was by far the favourite to become treasury secretary under Grant. Because he was close to Grant, he was close

to the Republicans, and he is America's preeminent financier. So it made perfect sense. He could have been like Hank Paulson or something. And so why didn't he? He didn't get the job. Okay, that must be a big blow to his ego.

Probably a huge blow. This is the guy that basically has succeeded and managed to do everything he's ever wanted. So at that point, he says, right, okay, I've got to prove that I am worth even more than they thought. And it was a big mistake to turn me down. Yep. So he throws himself into this Northern Pacific Railway. He'd sold a few railway

30:11 bonds in that boom before. So he underwrote the Lake Superior and Mississippi Railroad. And that had been like the bonds he sold had been 10 times over subscribers. And people wanted to buy 10 times more bonds than were on sale. That kind of really also made him realise, my God, we're leaving money on the table here. And he throws himself into Northern Pacific. So he strikes this fateful deal with this railway company.

And frankly, even calling it a railway company is a little bit fanciful. I mean, it's a shell, right? It's an idea. But he agrees that he's going to sell \$100 million worth of his bonds at a 7.3% interest a year. He's going to buy

them off the company at a 12% discount. So he can make that spread. If he can sell them at that interest rate, he makes

that 12% spread. And he was going to advance them and give them some of his own money, invest in the company itself. And in return, he was going to get an equity stake of up to 60% of the company. So this is him really branching out into becoming an industrialist, potentially himself. So he's massively exposed. I mean, it's a bit like putting all of your chips on the table onto black at a casino or something like that, saying, yes, I'm going to put it all into railways and basically, hopefully make a fortune as it goes right.

But raising money at that rate would have been quite high in those days. I mean, was a company that desperate or was it so risky that you had to pay a lot to get investors to get involved? Yes, essentially. I mean, this was one of the riskier projects out there. So some of the other projects that he had financed or helped finance before had been some more stronger cases. 7.3% was high, not egregiously high. It was higher

than what you maybe get for established railway lines where they're actually making money. But still, it's pretty low for something that is inherently still basically an idea. And I think he thinks he's not taking a huge risk. He's only promising to lend a few hundred thousand, \$500,000 of his own money in advance, and he's fabulously wealthy. And he's only promising to sell these bonds or try to sell them. And if he can, he makes a lot of money.

And if he doesn't, well, then it's fine. I mean, in return, he gets 60% of the companies he probably thinks. And it could still have been a dumb idea, but not an epic disaster that it ended up being. So basically, so he sets out and tries to build this railway, which goes through all these places like Dakota and Montana and places like that, which are pretty remote. You would have had a lot of Native Americans in that area as well, wouldn't you? I mean, what do they think about the idea? Well, they didn't like it.

Funny that. Yes, I know. I mean, this is one of the great tragedies because this was a man that in many ways for his era had reasonably enlightened views. He wasn't abolitionist, even though he was clearly not a super enlightened person. He was a great friend or saw himself as a great friend of many of the Native American tribes. He'd always respected them. But the Northern Pacific Railway goes

33:07 straight through a lot of crucial Native American lands and just complete north of the Iron Horse, as it was called. And, you know, that inevitably gets violent. So it's always expensive and difficult to build these railways. But this becomes basically like a little war. Like a Western movie with lots of, you know, you see those wonderful scenes, the railways going along and basically Native American groups attacking it or, you know, cantering along next to it on their horses and things like that. And the terrain, would that have been quite difficult? Because,

you know, it's pretty remote, pretty boggy, pretty tough to cross. Yeah, lethal and expensive and arduous and difficult. And I think they probably underestimated just how much it would cost and how long it would take. But the big problem was, it seems, these Native American attacks. Very understandably so. He uses government connections to get military protection for the construction. And, you know, it didn't help.

Like at times, you know, they were completely chased out of big areas of construction. They, you can read the annual reports where they talk because American companies started issuing annual reports around then. You can read about how they were essentially just not able to build in certain areas because of attacks from the local tribes. And animals as well, I imagine. You've got these buffalos wandering around and stuff. Well, one of the amazing things with American trains is that trains in America evolved differently from European ones. They got these massive bulbous fronts, heavy fronts with

basically plows. So you could drive at full speed and just like smash any bison and pulverize it. Wow. So you're dealing with bison, bogs, battles. You know, this is not the normal stuff of bond markets, is it? No, exactly. It is. He's essentially gone

all in on what ends up being a massive financial sinkhole, a black hole of money, essentially. But surely if he actually had a 60% equity stake, he would have controlled the company. Couldn't he have just said, listen guys, let's try and reroute a bit or try and, you know, deal with this in a more intelligent way? Well, one of the bizarre aspects of this is that although he was able to get up to 60% of the company, he didn't control the board. He didn't have control of the company. Why not?

We don't know. A blind spot. I guess he just thought, well, I don't want to stay involved in the nitty gritty. I'm the finance guy. I want to own this. I want all the upside and more of the upside, but I don't really want to deal with this. And unfortunately,

it proved that a lot of the Northern Pacific executives and the board were pretty incompetent. They weren't very good at building a transcontinental railway line. And he then found out through his horror that he really didn't have a chance to change the company where he thought necessary. I mean, was it possible to sell the bonds on the back of all this? He couldn't, in fact, sell most of the bonds. And that had led to two problems. Some of the bonds he had underwritten and put on his bank's

balance sheet, where they languished unsold. And an advance he'd promised the company so they could build construction, begin construction. He promised \$500,000, a big sum of money at the time, but something that he could easily carry. But that advance swelled to \$7 million. So he just kind of kept doubling down and doubling down as he was unable to sell the company's bonds. And frankly, just got deeper and deeper in the hole.

36:21 So what point did investors wake up and start to panic and refuse to buy the bonds? Arguably the real sort of moment. And this shows how finance is in the 19th century becoming a more global affair. It was actually a financial crisis in Austria of all places. Oh, wow. So it's back to that old interconnected world. The flap of a butterfly's wing

in one corner of the globe can cause a hurricane in another. Yeah. And this was an Austrian butterfly flapping its wings very violently. So Austria was enjoying its own financial boom at the time. There were lots of new companies, and this is very strong parallels with today. It was called the Gründerzeit, or the age of entrepreneurship. So lots of entrepreneurs were starting new companies, taking march of new

technologies in Austria, and the stocks were going high and the Austrian stock market was flying until it all basically imploded in May 1873. And it became known as the Gründerkrach, the entrepreneurial crash, essentially. Gründerkrach. Probably sounds better with a German accent. Yes. No, I'm not even going to try that. My German is very feeble these days.

And unfortunately, again, because finance is connected, it's global, a lot of the money that went poof in Austria actually came from Germans and Dutch merchants and investors and European investors in general, even some UK ones. And once they basically saw their money disintegrate in Austria, they started liquidating investments elsewhere. And European investors were huge in the American bond market. But in terms of what happened then, the European investors panic because of what's happening in Austria. They start to pull their money out of the American bond market. And then what happens?

His bank, Jake Cook & Co, is still one of America's preeminent banks. And they have other business and other stuff they do. But their balance sheet, they are sitting personally on a lot of Northern Pacific bonds they have not been able to sell. So they are running perilously low on liquidity, on just cash. And the US is still a pretty agricultural economy. And actually, Wall Street's cycles at the time were still very much basically set by the agricultural cycles, which is why people often withdrew money when there was planting season and harvesting season.

And that's why by September, by that September, I mean, the bank had quietly almost come close to failing in 1872. And that only emerged later. And it was kind of bailed out by the fact that the American government knew that there were these pressures every autumn. So they kind of pumped more monies of what we call

today QE, but then make sure there was enough sort of free collateral and money coursing around the system in 72. But in 73, it's just so much more acute because of this issue in Europe. But if Jay Cook was so close to Ulysses Grant, and if this bank and this project was so central to the American financial markets, if not the economy, why didn't the government just step in to bail it out?

39:28 We don't know. But we do know that Ulysses Grant, the president, visited Jay Cook just days before the bank finally had to close its doors. And does anyone know what they talked about? No. We know that Grant went there, I think it was three days before the big crash. So 15th of September 1873, Grant visited Jay Cook in August and stayed overnight on his way to visit his son at a nearby university. Nobody knows what they talked about.

But we do know that the government did not step in. And in fact, arguably, I mean, even if they don't bail out Jay Cook themselves, but they could have made sure that there's enough money sloshing around the system in New York. Well, if this was a movie, that scene of them meeting in that overnight stay would have been an extraordinary cliffhanger moment. Because of course, then what happens next is it goes bust. Yeah. Tell us what happened and why it turned into the Lehman Brothers of the late 19th century.

So Jay Cook and co had three main branches in the Washington, New York, and Philadelphia. And Jay Cook mostly worked out of Philadelphia, but it was in New York a fair bit. But New York, he had his own partners that were very able, but they literally just ran out of money. This wasn't necessarily, though I think it proved to be a solvency issue, but it was a liquidity issue. They didn't have enough money to meet their own obligations. People pulling out money because of these issues we talked about, and they didn't have enough money at hand.

And so the New York partners summon all the big counterparties among the big brokerage houses and investment banks on Wall Street and say, look, if we don't have a million dollars by 10 a.m., we're dead. Wow. Yeah. That's where you get the spooky music and dramatic music into the movie. It is the moment really when, you know, it electrified everybody. This again, this was not a small bank. This was like a JP Morgan or a Goldman Sachs out of the blue suddenly

saying we're dead. I think people knew that Jay Cook and Co. might have been in trouble or had problems with bonds and these Northern Pacific bonds, but nobody really understood or appreciated the depth that this bank was teetering on failure and was now basically dead unless everybody else stumped up a collective bailout. And they weren't able to do that. So they had to literally just shut the doors. That's how they closed banks. They shut the doors. And then the New York partners telegraphed

Jay Cook in Philadelphia to let him know what they've done. There's actually a report here from how he reacted, which I was going to say, did he show his face or. Well so here it is. He read the dispatch announcing the suspension of Wall Street, sorrowfully ordering the doors of his third street house to be closed also. And then as he turned his face away from the men who surrounded him, the tears streamed from his eyes. It was an unusual sight. No one in his office has seen this great strong man, the pillar of a nation

42:23 weep, but it was soon over. So that's a quote from Ellis Paxson Oberholzer who wrote the earliest biography of Jay Cook in 1907 on the closure of the Philadelphia branch. A bit like imagining Jamie Dimon or Larry Fink, um, suddenly bursting into tears in front of everyone on, you know, live TV or something like that. Or Lehman Brothers CEO, Dick Fulda. I don't think I ever saw him cry, but, uh, he doesn't look like the crying type either, but neither was Jay Cook until he finally had to make, do this, take this step. What happened afterwards? Did Jay Cook lose everything? Did he have to move out of his

fancy estates and go and live in penury? Did he go and run away to Mexico? What happened? Well he had to declare bankruptcy. The bank was bankrupt. So was he personally. Uh, he

lost this incredibly opulent estate that he had built and financed himself. It was, I had a Roman, fake Roman ruins. It had his own personal telegraph station, swimming pools, its own personal 30. It was just frankly one of the most opulent estates in America at the time. And he lost it. He had to move in with his daughter.

Wow. And did he stay penniless then throughout his life? Or become a monk or anything like that? No, he didn't. He did not die a poor man though. He didn't die. Maybe as obscenely

wealthy as he'd been. Uh, by 1880 he'd actually regained a lot of his money. Uh, he'd invested in a Utah silver mine alongside Jay Gould, the, the railway robber baron. Wow. And so he actually stayed a rebound. Yep. So modestly so. Uh, so he was able to repurchase organs, his estate, but it was

so expensive maintain. He actually donated it and turned into a school for girls. And is it still standing? I believe it is. Yes. Though in, in, you know, very different shape.

So that was a story of Jay Cook. And what about the American economy as a whole and the financial system? Did that rebound or stay in the doldrums for a long time? Yes. So it took a long time to recover. It did recover. The US economy always does. Uh,

but it took six long years. It was a massive crash. It rippled everywhere. It hit Europe, it hit the UK. But for a long time, it was actually called the great depression until we actually had the even greater depression in the thirties. And now economic historians often call it the long depression just because of the length and severity of it.

And so what happened to all the other railway companies? Did they all topple over as well? Most of them did. As one railway baron once said that, you know, no railway is really economically feasible until it's gone through bankruptcy at least once. And that proved to be true. Uh, by the end of 1875, there were over 120 railways with a \$550 million worth of bonds gone bust. Half of all US iron foundries closed by the end of 1874 because with all these railways

45:12 bust, there was less demand for iron. Corporate bankruptcies doubled, uh, from 1873 to a peak of 10,000 bankruptcies in 1878. Uh, unemployment soared, uh, deflation set in. This was, you know, a period of hard money. There was still the gold standard. So there was limited what the government could do. Uh, there was no central bank that could lower interest rates really.

Um, words like bum, uh, became sort of more common after this to describe, uh, people that were homeless and living destitute. So you really had a sort of libertarians dream of a hard money shake out or a kind of creative destruction period. But did anything come back later on? I mean, was that the end of railway bond booms in America or did it arise again? Nope. They, you know, within a decade though, uh, booming again, because I mean, this was a, a transformational technology. Again, we, we underestimate just how powerful it was.

There's this great, uh, line in a poem by Joaquin, uh, river, uh, frontiersman poet in America. So there's more glory in the rush of a single railway track than there was in the sacking of Troy. Lots of towns still needed railways. So the boom revved up again. And again, like lots of those numbers we had from 1890 shows just how big it was. America was basically then soon on its way to having more railways and almost the rest of the world combined. So it's that eternal American optimism

or that Silicon Valley mentality of that failure is not a badge of shame. It's an incentive to try again. Yes. Failure is temporary graft and grit and determination and just pushing ahead. That's forever. Yeah. Was there another railway bond crash or was that the last one? There've been

several since then. And then the UK and then in the U S that periodic financial crisis, but nothing of that severity until 1907, uh, that was more generalized financial crash. And as you and I know that actually helped spark the birth of the Federal Reserve because it was such a big crash. Well, we're going to come back to that in another episode because that's another amazing story. But, um, this is just a made for movies tale about

booms, busts and another boom again. Yes. I do wonder who would play Jay Cook in a movie though. Daniel Day-Lewis? He'd do a phenomenal role.

That's kind of intensity. I think so. Yeah. But you need to grow a big bushy Santa Claus beard. Oh, he could do that. I'm sure he could, but yes, I think he'd have that kind of glowering presence on screen to kind of capture all the contradictions. Okay. So let's step back

for a moment and think about what the lessons are from the financial history just described for the present day. One of the interesting questions to think about is are bubbles bad or not? Because, you know, I read a fantastic book a few years ago by the author Daniel Grose about called Pop, which basically argues that when you get a crazy bubble and amount of money pouring into a sector, that's completely insane under some measures. What you do is you create an infrastructure that not all the investors funding that infrastructure will benefit from, but society as a whole will be left with something, even after the bubble pops, that actually is of use and helps to pave the way for more growth and innovation in the future.

48:34 You know, so we saw that, say, with the fiber optic cables, with the whole, you know, telecoms boom and IT boom in the late 1990s. You know, we were left with what essentially became the infrastructure for the internet. Is that what happened with the railway boom? I mean, the fact is that when the bubble burst, the railways were still there, weren't they? I mean, manias and bubbles and crashes, they get a bad name, especially when they cause economic crises. But the reality is, and this is hard to say, because obviously economic crises are painful and people lose their jobs. Literally, we know people die more in depressions. But the optimal

number of manias is definitely not zero. Manias is kind of, we can see it's kind of hard wire into our psyche. We can see them across so many countries, across so many times. And very often, I'd say more often than not, they do have positive side effects or direct effects that after the crash, what we are left with is, yeah, fiber optic cable. We cable the world. Railways that knitted the US together, turned it into the world's most powerful economy. Canals as well, telegraphs.

There's so many real estate examples. And Florida was a swampy hellhole until somebody decided, oh, we're going to sell it as a nice place for retirees. HANNAH-JONES So, I mean, some of today's modern big tech leaders like Jeff Bezos have said, well, if you look at that parallel for the modern world, the AI boom is also in some ways crazy and hyped up. And at some point, there's probably going to be, you know, a bubbling, bubble bursting there. But that will also leave behind an AI infrastructure, which will help the world. So, he says it's actually a good bubble, quotes, as opposed to something like the 2007 or 2005, 6, 7, subprime bubble, which wasn't such a good bubble. Would you agree?

MARK Well, I'd say, I mean, 07, 08, you know, there was lots of real estate being built. The US probably, you could argue, has a cheaper housing stock for ordinary people now than it had because of that building boom. I think with AI, it depends on like how valuable the overall infrastructure is, because you can get left with this massive series of white house. Do we need these vast data centers the size of Manhattan to create like fake videos of cats? Is that really something that's going to usher in the next era? Possibly not. But I am hopeful that actually, yes,

clearly there is a bubble. Everybody, even people inside, it's very rare that you hear people inside those industries say very honestly, oh, yes, it's definitely a bubble. But I am very optimistic that we will be left with good stuff after it. I mean, just on the energy infrastructure, they're going to have to build so much more power generation capacity to power these data centers. And that could be a huge boon. HELEN Absolutely. I mean,

51:22 I guess one of the differences that worries me is that the railway lines, the railway track that was laid down in the 1870s and beyond, is still physically sitting there in many cases, 150 years later. Whereas it's not clear to me that the data centers will be as durable, not just because chips tend to have a half life, but also because everyone right now is betting on a certain form of AI, essentially large language models. And there

are other types of AI coming up, you know, around world models, and your symbolic AI, which haven't got as much attention, but might actually end up supplanting some of the other techniques. And so you could end up with a situation a bit like VHS and Betamax, where you have essentially stranded assets.

MARK Yeah, no, I mean, not all capex booms are created equal, unfortunately. And there is a danger that by pouring all our efforts and money and attention into one area, then something else is starved of attention. And you could argue there's all sorts of things like renewable energy that arguably could maybe some of the hundreds of billions of dollars now being spent and lent towards AI data centers be more fruitfully diverted elsewhere. That's what we don't really know. And I think in a capitalist economy in a capitalist system, you know, it's, you know, there are stupid things that happen all the time. It's just very hard to know how stupid they are and what is stupid before the fact. But you know, I look forward to doing this podcast in 50 years time, we can find out just how

durable some of these things ended up being. Well, who knows if they'll still be podcasting 50 years time. Yeah, there's a reason why people say the foremost dangerous words of finance is this time is different. Or as I say, those who do not study history are condemned to repeat it and lose money. Yeah. So that's it from us this week. You've been listening to the story of money

from the Financial Times with me, Gillian Tett. And me, Robin Wigglesworth. And we'll meet again here next week. See you then.

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